ThyroPredict AI

Comprehensive Project Overview, Technical Architecture & Clinical Features

Production URL: https://thyropredict-ai.vercel.app | **Version:** 1.0.0 Production Release | **Date:** October 2026
Repository: https://github.com/nohinme-ux/ThyroPredict-AI.git | **Lead Researcher Account:** researcher@thyropredict.ai

> ***Clinical Classification:*** *Explainable AI (XAI) Clinical Decision Support System (CDSS) for Thyroid Spectrum Disorders. Designed for clinical audit, endocrinological research, and reproducible biostatistical modeling.*

1. Executive Summary & Clinical Rationale

Thyroid disorders—predominantly autoimmune Hashimoto's thyroiditis, primary hypothyroidism, and thyrotoxicosis—represent some of the most prevalent endocrine dysfunctions worldwide. While routine blood tests evaluate serum levels of Thyroid Stimulating Hormone (TSH) and Free Thyroxine (FT4), accurate differential diagnosis often requires complex multi-parametric synthesis across autoantibodies (Anti-TPO), inflammatory cytokines (IL-6, Visfatin), lipid profiles, and metabolic comorbidities.

Modern deep learning architectures (e.g., deep neural networks, ensemble gradient boosters) frequently attain high benchmark accuracy but operate as opaque 'black boxes.' In clinical healthcare, an unexplainable diagnosis creates severe medico-legal liabilities and fails to gain physician trust. **ThyroPredict AI solves this dilemma** by engineering deterministic, white-box **Decision Tree classifiers**. Every diagnostic prediction is coupled with an exact, node-by-node pathophysiological explanation rule traversing verified biochemical thresholds.

2. System Architecture & How It Works

The application is structured into four tightly integrated layers, deployed on a unified global serverless cloud topology:

- **1. Clinical Biomarker Ingestion:** A responsive React 19 single-page interface collects 23 standardized clinical, hemodynamic, and biochemical biomarkers with client-side range validation.
- **2. Strict Pydantic Data Clamping:** The FastAPI backend validates physiological bounds (e.g., TSH between 0.0001 and 300.0 uIU/mL, Anti-TPO between 0.0 and 2000.0 IU/mL), neutralizing malformed payloads and injection attempts.
- **3. Dual-Runtime Decision Tree Traversal:** The system evaluates patient feature vectors against pre-trained decision tree graphs, outputting the predicted diagnostic class, calibrated confidence scores, and exact decision path rules.
- **4. Relational Persistence & Audit Trail:** Predictions are linked to anonymized participant profiles in a serverless relational database, immediately populating real-time analytics dashboards and longitudinal audit logs.

Layer	Component	Technology	Key Functional Responsibility
Client Frontend	Web Application SPA	React 19, TypeScript, Vite	Interactive forms, visualizer, audit drawers, KPI charts
API Gateway	Serverless Endpoint	FastAPI, Starlette, Vercel	Sub-second REST routing, JWT middleware, CORS negotiation
Machine Learning	Dual Decision Trees	Scikit-learn, Native Graph	Zero-dependency edge inference, deterministic rule extraction

Data & Storage	Relational ORM	SQLAlchemy 2.0, SQLite	Participant records, prediction histories, consent tracking

3. Dual Machine Learning Model Specifications

ThyroPredict AI incorporates two distinct models trained and audited on rigorous clinical study cohorts:

Specification	STS Clinical Model (Primary)	UCI Benchmark Model (Validation)
Dataset Source	STS Clinical Thyroid Study (269 patients)	UCI Thyroid Disease Repository (972 patients)
Biomarker Inputs	23 clinical, lipid, hormonal, & serology markers	21 standardized clinical & demographic features
Diagnostic Classes	Autoimmune Disease, Euthyroid, Hyperthyroid, Primary Hypothyroidism	Negative (Euthyroid), Primary Hypothyroid, Compensated Hypothyroid
Test Accuracy	**94.44%** (Balanced Accuracy: 94.87%)	**99.07%** (Test Accuracy)
Macro F1-Score	**91.98%** (CV Macro F1: 1.00 ± 0.00)	**94.61%** (Macro F1-Score)
Primary Discriminators	Anti-TPO (> 37.50), TSH (> 5.16), FT3 (<= 2.15)	TSH (> 6.05), FTI (<= 64.50), TT4 (<= 59.00)

4. Key Application Features & Functions

Feature 1: Interactive Clinical Predictor (/predict): Empowers researchers to input multi-parametric clinical profiles. Includes a real-time model switch (STS vs. UCI), instant physiological sanity validation, and four one-click clinical scenario presets: Hashimoto's Autoimmune Disease, Overt Primary Hypothyroidism, Euthyroid Control, and Thyrotoxicosis.

Feature 2: Explainable AI Decision Rule Visualizer: Deconstructs every diagnosis into transparent tree traversals. Clinicians can review the exact biomarker threshold that triggered the branch, inspecting the leaf purity, sample support, and multi-class probability breakdown.

Feature 3: Research Participant & Cohort Registry (/participants): Maintains longitudinal subject tracking with unique de-identified study codes (e.g., THYRO-PART-001). Tracks consent status, demographics, clinical notes, and maintains cascade referential integrity across prediction evaluations.

Feature 4: Real-Time Analytics & Cohort Overview Dashboard (/): Presents high-level research KPIs: total cohort enrollment, cumulative predictions executed, diagnosis prevalence pie/bar distributions, and a live stream of recent diagnostic transactions.

Feature 5: Longitudinal Prediction History & Audit Drawer (/history): A searchable, filterable audit registry. Researchers can search by patient code or diagnosis, open an audit drawer to inspect the original raw 23-biomarker payload, review previous decision rules, and export specific cases.

Feature 6: Dataset Insights & Exploratory Data Analysis (/dataset-insights): Visualizes the clinical distribution of the underlying training data, providing box plots, quartiles, and epidemiological breakdowns for TSH, FT3, FT4, and autoantibody titers across cohorts.

Feature 7: De-Identified Research Data Export (/api/research/export): Generates an RFC-4180 compliant CSV stream containing complete biomarker vectors, predicted outcomes, confidence scores, and decision path strings formatted for instant biostatistical analysis in R, SPSS, Stata, or Python.

Feature 8: Role-Based Authentication & Cryptographic Security: Secures all clinical endpoints with cryptographic JWT tokens and salted bcrypt password hashing, ensuring that sensitive study data remains fully protected in multi-tenant environments.

5. Complete REST API Reference Table

Method	Endpoint Path	Auth Required	Description & Return Payloads
GET	/health	Public	Service health check and model loading status verification
POST	/api/auth/login	Public	Authenticates email/password; returns JWT access token
POST	/api/auth/register	Public	Registers a new clinical researcher user account
GET	/api/auth/me	Bearer Token	Retrieves profile information for the authenticated user
GET	/api/model/info	Public	Returns active model catalog, feature counts, and versions
GET	/api/model/metrics	Public	Returns accuracy, confusion matrices, and feature importances
POST	/api/predictions	Bearer Token	Executes decision tree inference; returns class & rule paths
GET	/api/predictions	Bearer Token	Retrieves historical predictions log with filter parameters
GET	/api/predictions/{id}	Bearer Token	Retrieves detailed prediction record and raw feature payload
DELETE	/api/predictions/{id}	Bearer Token	Permanently deletes an individual historical prediction record
GET	/api/participants	Bearer Token	Lists all enrolled research cohort participants
POST	/api/participants	Bearer Token	Creates a new research participant with consent status
GET	/api/overview/stats	Bearer Token	Aggregates dashboard metrics, KPI counters, and breakdowns
GET	/api/dataset/summary	Bearer Token	Descriptive statistics and demographic summaries of datasets
GET	/api/research/export	Bearer Token	Streams complete de-identified audit trail as an RFC-4180 CSV

6. Research Use & Regulatory Disclaimer

Notice of Research Use Only (RUO): *ThyroPredict AI is engineered solely as an academic and clinical research decision-support platform. It is not approved by the U.S. Food and Drug Administration (FDA), European Medicines Agency (EMA), or Central Drugs Standard Control Organisation (CDSCO) as an autonomous primary diagnostic device. Healthcare practitioners must integrate ThyroPredict AI predictions with independent laboratory serology, ultrasonography, clinical symptomatology, and medical history before instituting clinical management or pharmacotherapy.*